

# DSS5105 Assignment-1

## SG Private Housing Transaction Prediction

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## 1 Analysis

### 1.1 Top 3 Key Factors

Based on the SHAP (SHapley Additive exPlanations) analysis of the optimized Gradient Boosting Tree (GBT) model, the following three factors were found to have the highest mean absolute contribution to the transaction price:

- **Area (SQFT):** SHAP importance score of 0.261558. Property size shows the strongest intuitive relationship with price, confirming that Area is the primary fundamental driver. Larger units directly correlate with higher transaction values, reflecting Singapore's premium on space.
- **Project Name (Encoded):** SHAP importance score of 0.123095. This factor captures the residual price premium associated with specific developer reputation and amenity levels, after accounting for size and general location. It signifies the market's trust in established brands and quality assurance.
- **Type of Sale (Resale):** SHAP importance score of 0.088324. This factor segments the market based on transaction type, reflecting the distinct pricing dynamics between new launches (primary market) and resale properties (secondary market), often reflecting market conditions and property age factors.

### 1.2 Patterns Trends

- **Location premium:** Prime districts (9-11) command 150-200% premiums over outer districts (23-27), demonstrating extreme location-based pricing tiers [Section 3.1].
- **Seasonal effects:** November shows the highest median prices and peak transaction volumes, with Q4 commanding 8-12% seasonal premiums over other quarters [Section 3.4].

- **Tenure impact:** Long-term leasehold comprises 59.68% of properties, with freehold commanding 20-30% premiums over short-lease properties [Section 3.3].
- **Size correlation:** Larger units show exponential price increases, indicating the concentration of the luxury market [Section 2.4].

### 1.3 Buyer Recommendations

- **Districts:** Focus on Districts such as 17, 25 offering average SGD 1.1M properties, representing 50%+ savings versus premium districts 9-10. [Section 3.1]
- **Purchase during low demand months:** Purchase during August, December, or January: Purchasing property in August/December/January is a good idea, as the mean and median property prices are the lowest in these months, likely owing to demand being low during these months (festive or school beginning). [Section 3.4]
- **Consider a smaller property size:** Intuitively smaller properties cost less; they can offer considerable savings over a period of time. [Section 2.4]